MILITARY SPECIFICATION

DIGITAL WORD GENERATOR PERFORMANCE AND INTERFACE

This specification is approved for use by Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the minimum performance characteristics for a commercially available Digital Word Generator (DWG). The DWG is intended for use in computer controlled test systems and specifically the hybrid test system described by MIL-H-85330(AS). The document provides sufficient information to an offeror of a hybrid test system to integrate the DWG into his system or to provide a fit, form and functionally compatible unit with the Navy's existing inventory of DWGs.

2. APPLICABLE DOCUMENTS

2.1 Publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

SPECIFICATIONS

MILITARY

MIL-H-85330(AS) - Hybrid Test System Specification

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD) Code 93, Lakehurst NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD for 1426) appearing at the end of this document or by letter.

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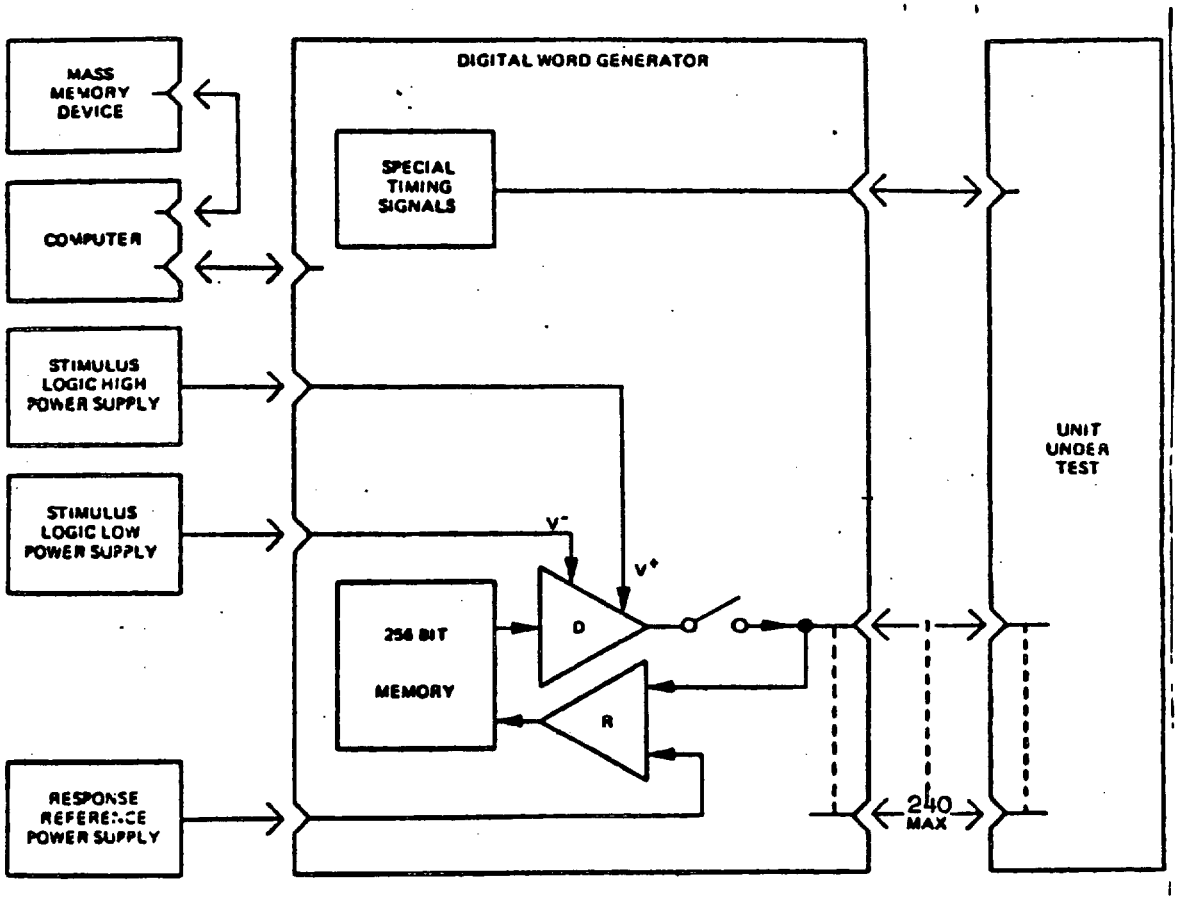


FIGURE 1. System Configuration.

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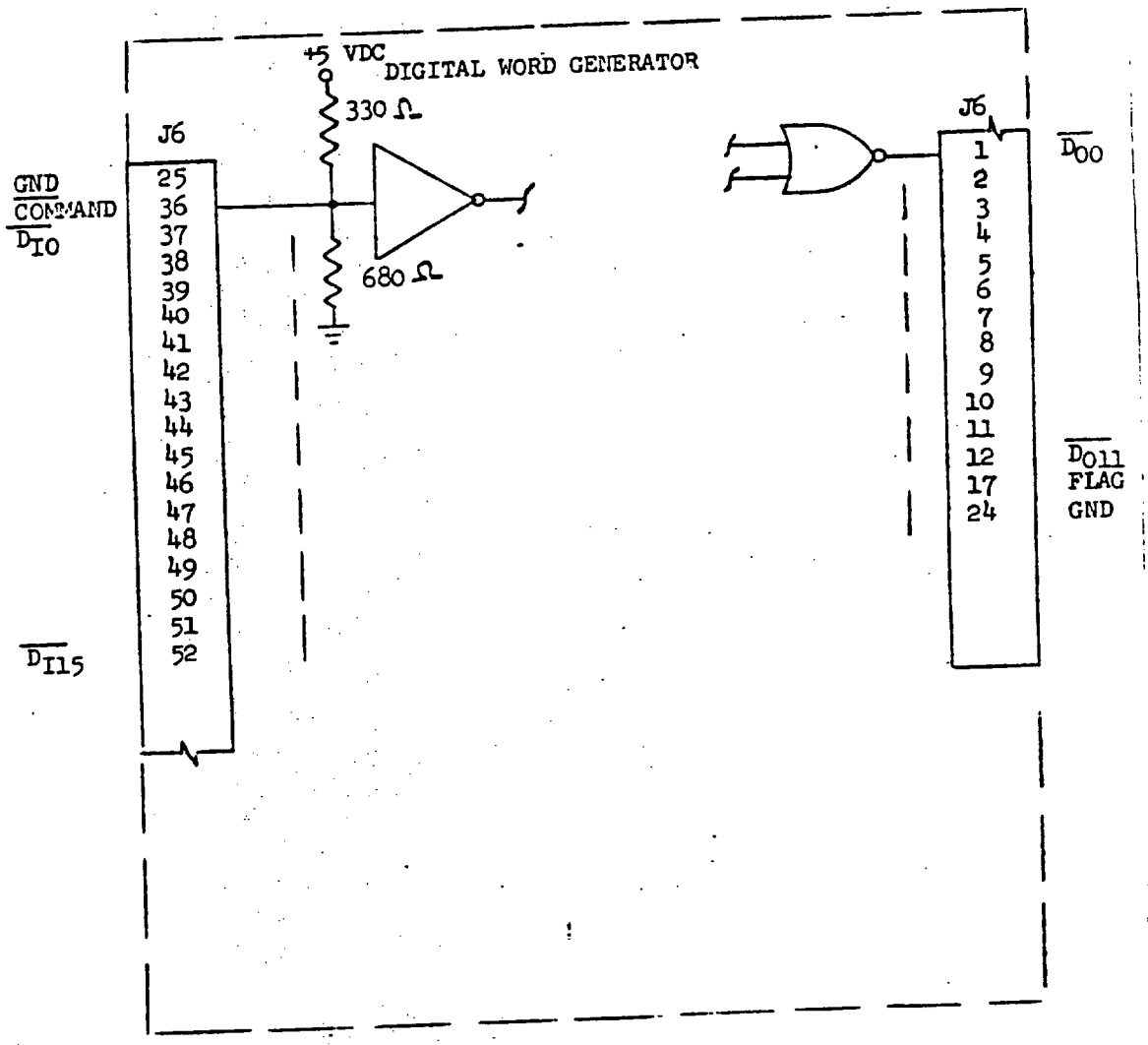


FIGURE 2. Computer/DWG I/O.

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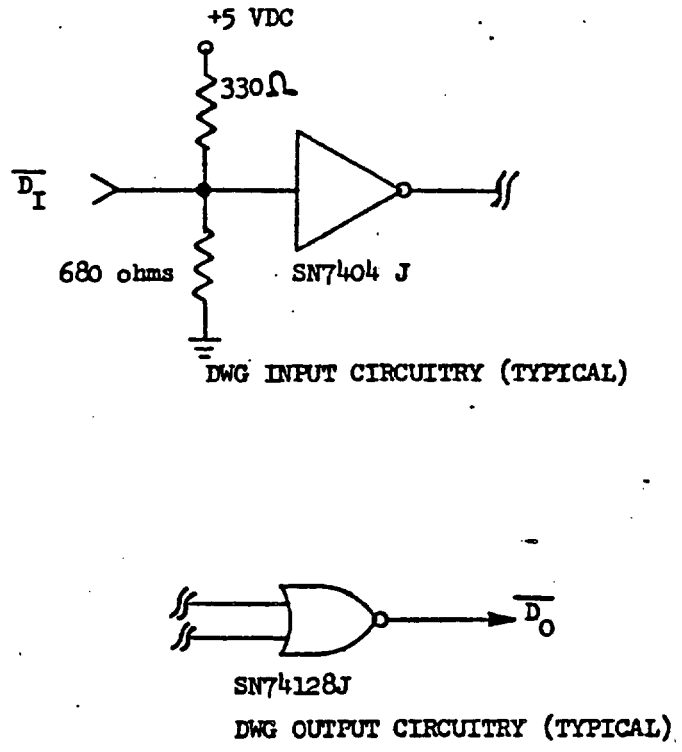


FIGURE 3. DWG/Computer interface.

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3. REQUIREMENTS

3.1 General. A typical system configuration is shown pictorially in Figure 1. The DWG shall be controlled by a digital computer which supplies 16 bit binary words to control the mode of operation of the system. Data and control signals are exchanged between the computer and DWG through connector J6, Figure 2, (connector information per DWG Outline Drawing) on the rear of the DWG. Computer inputs (16 data bits) and a control command line, to the DWG are routed to Control Card A1 (A31U30051). The Control Card is supplied with all DWGs and provides the proper terminations (Figure 3) for digital data sources employing Transistor Transistor Logic (TTL) microcircuit logic output drivers having ground true logic levels (Logical 1 = LO; Logical 0 = Hi). The DWG also outputs data to the computer (12 data lines and a control flag line) via DWG connector J6. The lines are driven by microcircuit output drivers located on the Control Card A1 within the DWG. The data lines and flag drivers produce ground true logic levels compatible with systems employing TTL logic circuitry. An example of a typical computer input/output card is the Hewlett Packard HP12566-60024 Microcircuit Interface Card.

3.2 Mechanical Packaging. The DWG[1] is packaged in one (1) drawer containing the following circuit cards:

a. Control Card (A1) - P/N A31U30051. The Control Card accepts data and instructions from the Computer in order to control both DWG memory and Timing Generator operation. All data to and from the Computer passes through this card.

b. Timing Generator Card (A12) - P/N A31U30052. The Timing Generator Card accepts instructions and data from the Control Card to generate the timing required for DWG memory access and UUT testing. Functionally, the Timing Generator generates the programmable test clock, sets the test length, loads and increments the test pattern counter, generates the stimulus ($\overline{S1}$, $\overline{S2}$, and $\overline{S3}$), response ($\overline{R}$), and write (WR) strobes for the DWG and UUT. It also provides for external timing and synchronization modes with the UUT.

NOTE 1/ A brief description of the DWG and its theory of operation is provided in an Appendix contained herein.

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c. Memory Card (A2 through A11) - P/N A31U30053. Each memory card contains 24 independent input/output lines which are available to generate stimuli and sample responses. Each card is sub-divided into two (2) halves: (1) Pins 1 to 12, and (2) Pins 13 to 24. The entire card and all of its functions are controlled by the Control and Timing Generator Cards.

3.3 Electrical Requirements

3.3.1 AC/DC Power Requirements. AC power is brought to the DWG via connector J2 to operate its internal blower. DC (+5V) power is brough in via connector J4. A maximum blower power of 15 watts and a maximum of 55 amps DC are required per drawer (240 pins). Figure 4 illustrates the method of applying power and shows pin assignments. The AC power, used to operate the blower (B1), is supplied by a detachable three (3) conductor power cable.

3.3.2 Logic Level References. All logic level reference voltages applied to the programmable drivers/receivers (located on Memory Cards) are supplied from external power supplies. Since the requirement for references varies according to the user's test application, the power supplies are not furnished as part of the DWG and must therefore, be supplied by the user.

3.3.2.1 Logic Level Reference Group 'A' and 'B'. When testing UUTs that are configured with different types of logic element (e.g. Metal Oxide Semi-conductor (MOS), Diode Transistor Logic (DTL), Resistor Transistor Logic (RTL), TTL, etc.), at least two (2) groups of programmable ($\pm$20V) logic level references are needed. Each group requires three (3) voltages to establish the analog threshold levels for driver Hi, driver Lo and references. Therefore, the six (6) reference voltages used for groups 'A' and 'B' are obtained from six (6) programmable power supplies external to the DWG.

3.3.2.2 Reference groups 'A' and 'B' are fed into the rear via J3. Figure 5 illustrates the method of inputting logic level reference groups 'A' and 'B' and distribution in the DWG. Although both 'A' and 'B' references enter the memory card, only one (1) of the references are used for that

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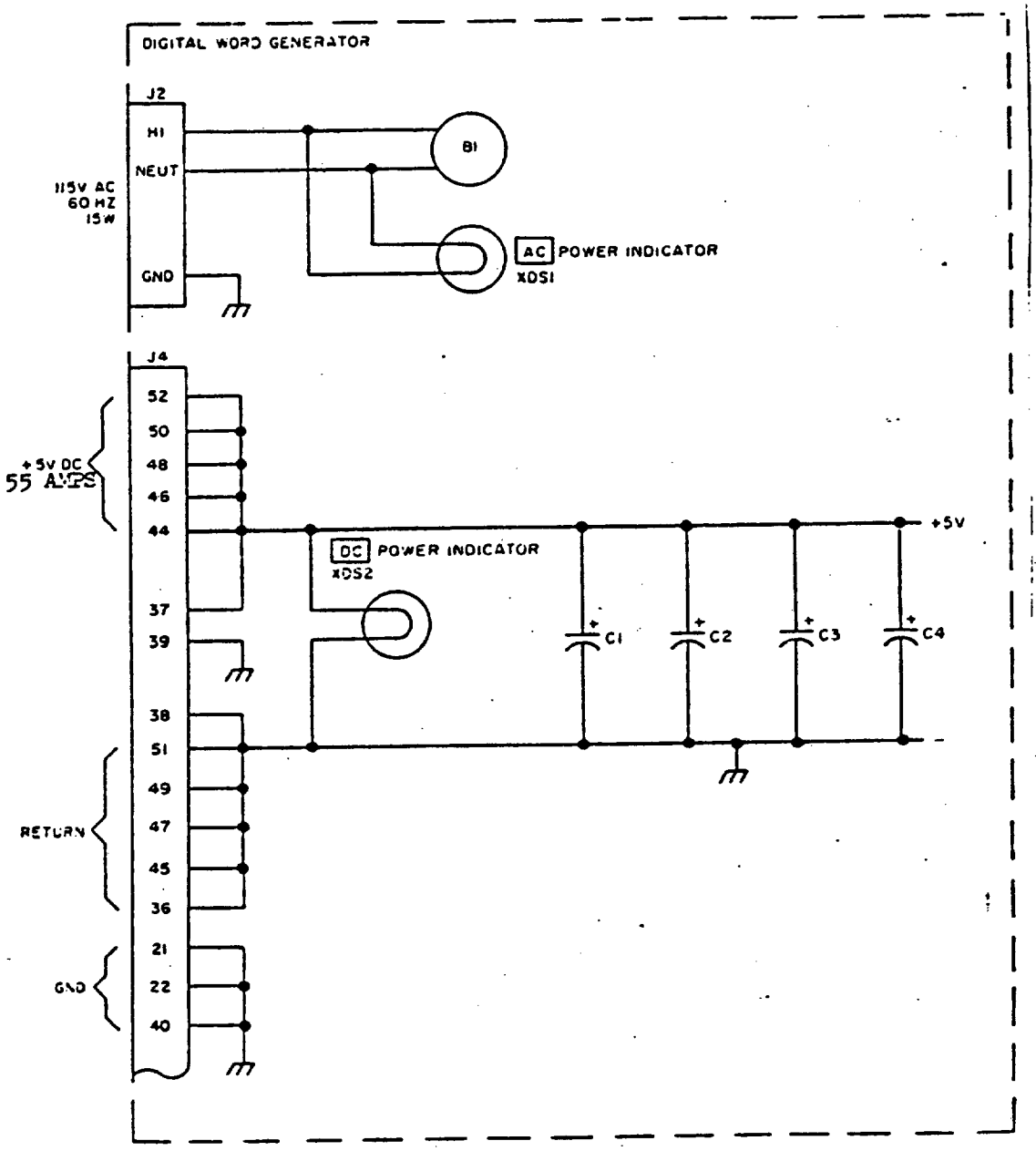


FIGURE 4. AC/DC power requirements.

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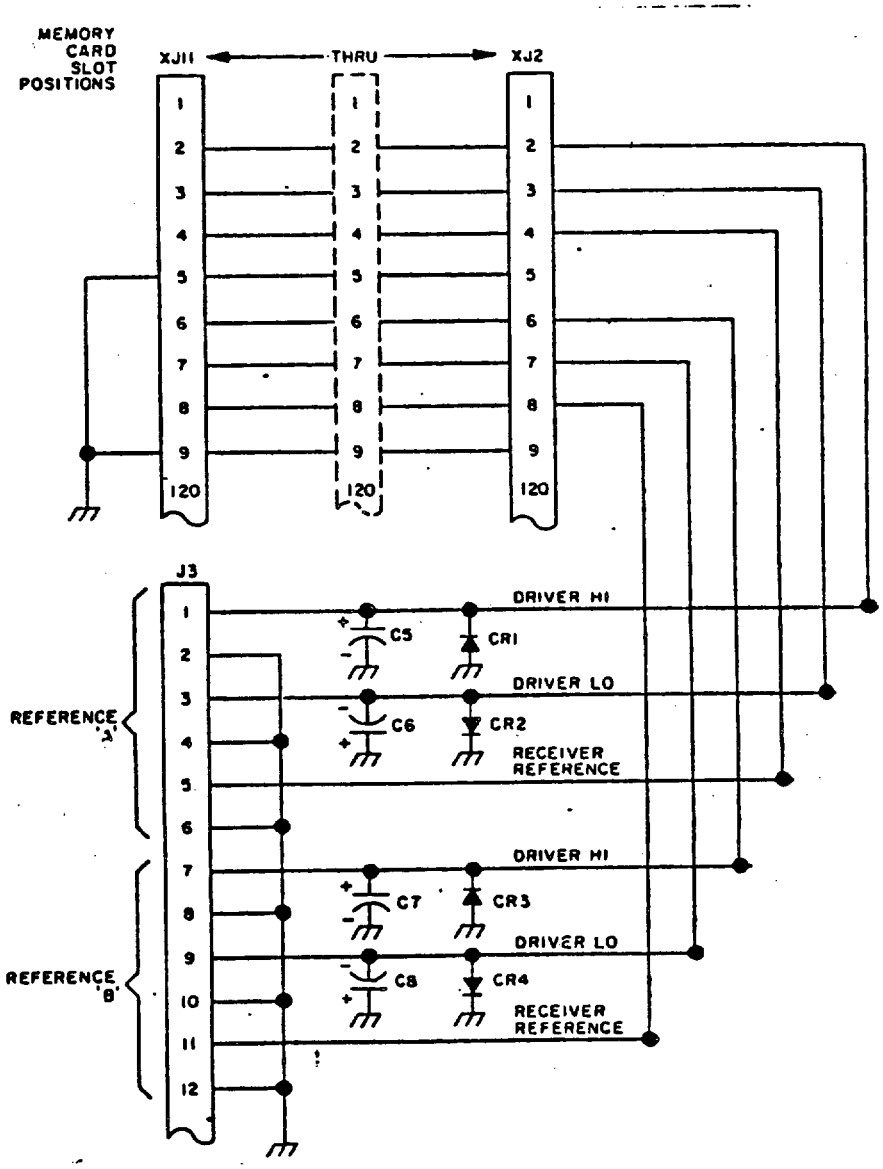


FIGURE 5. Logic level reference groups A and B input and distribution.

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card. The choice is made by adding three (3) jumpers to a jumper block (see Figure 6) located on the programmable memory card.

3.3.2.3 The DWG is configured such that pins 1-120 are assigned to reference group 'A' and pins 121-240 are assigned to reference group 'B'.

3.4 Signal Interfaces

3.4.1 DWG/UUT Input/Output. Memory Cards (1-10) provide the digital test signal interface with the UUT, in modular increments of 24 I/O test pins per card. Each test pin is programmable as either a driver '1' (for inputting test logic stimuli to the UUT) or receiver '0' (accepting test logic responses from the UUT). Ten (10) Memory Cards are installed in the DWG. This provides a digital test interface with the UUT of 240 I/O test pins. The 24 I/O test pins are made available via the edge connector (Figure 7) on the top of each memory card. Mating connector and information for interfacing with edge connector J1 is listed in Table I.

3.5 Digital Word Generator Performance. The digital word generator performance is in accordance with the following:

a. The unit shall provide capability for dynamic testing of units under test having up to 240 pins digital interface. Each test pin provided for this purpose shall be operable as either a driver or comparator under computer control.

b. The digital word generator shall provide a 256 bit memory for each test pin.

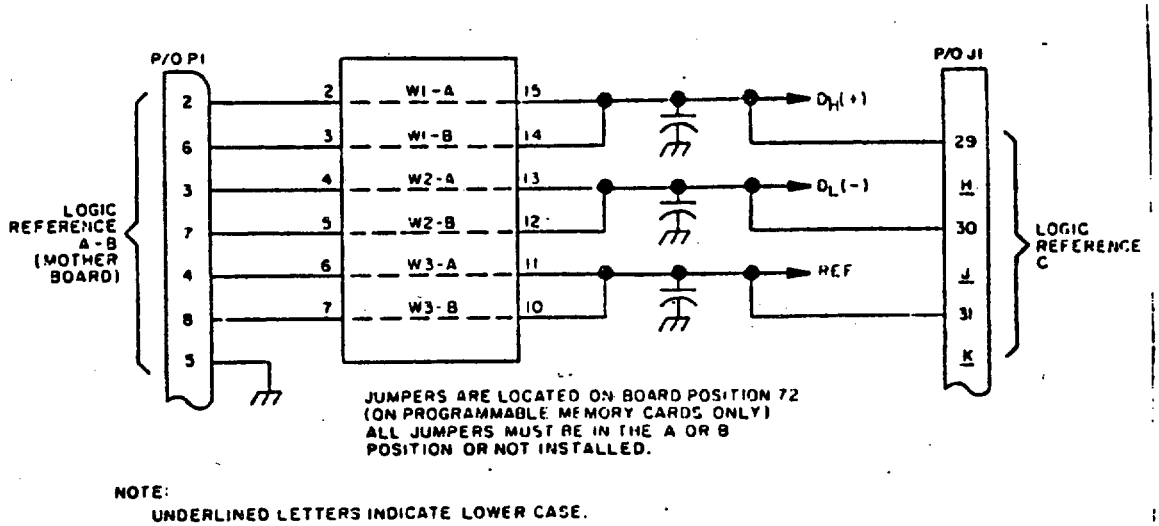
c. Logic levels for the 240 pins shall be programmable over the following voltage ranges:

1. Programmable Drivers. The high logic level (logic 1) shall be programmable from 0 volts to +20 volts. The low logic level (logic 0) shall be programmable from -20 volts to 0 volts. The maximum differential voltage (high minus low) shall be 20 volts.

2. Programmable Receivers. The logic level (threshold reference voltage) shall be programmable from -20 volts to +20 volts. The maximum signal input voltage shall not exceed $\pm$20 volts.

3. Driver/Receiver Programming. The levels shall be programmable in two (2) independent groups. Group 1 encompasses the first 120 pins, and Group 2 encompasses the second 120 pins. The programmable drivers shall source up to

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NOTE:
UNDERLINED LETTERS INDICATE LOWER CASE.

FIGURE 6. Logic level reference group selection.

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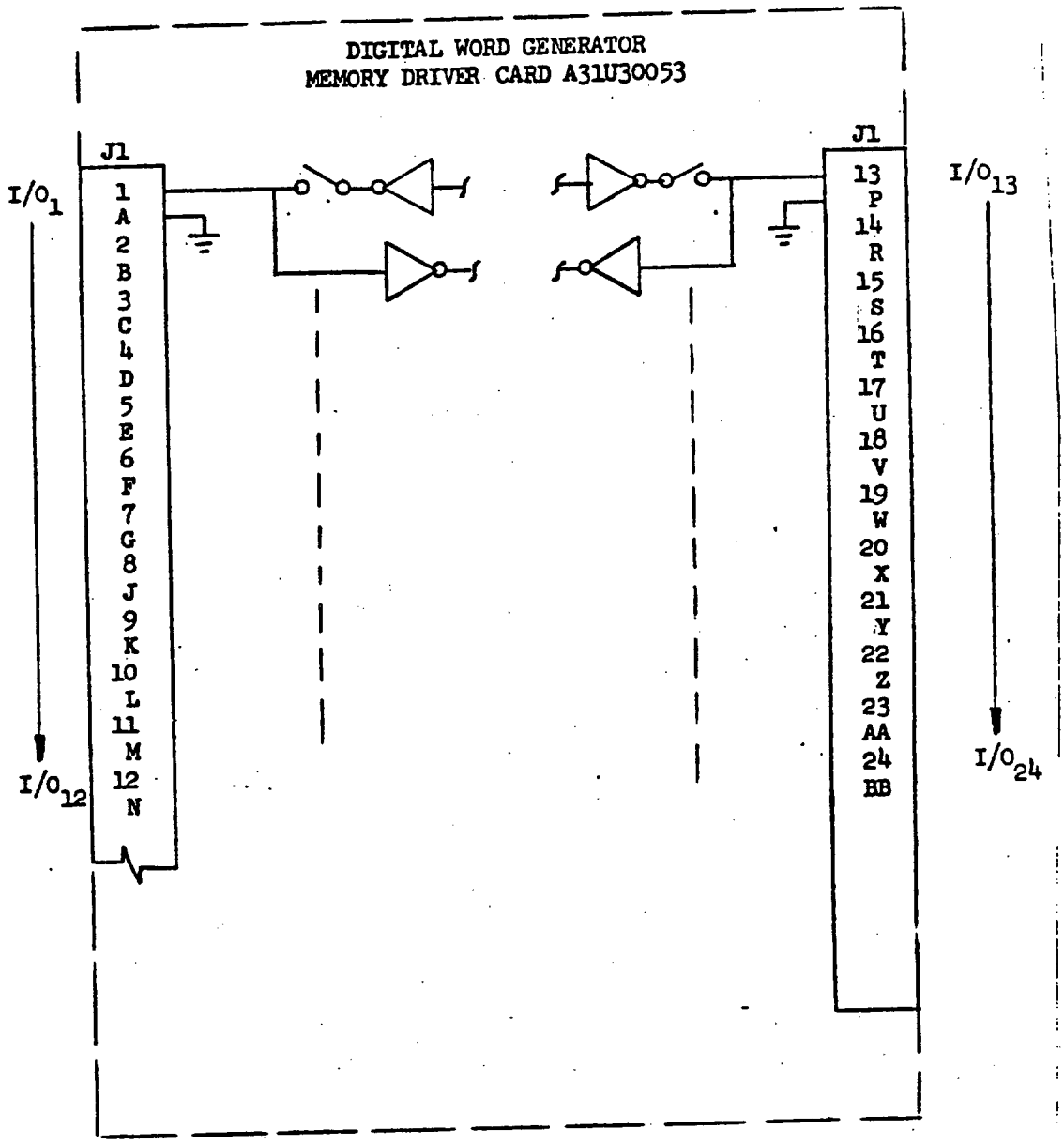


FIGURE 7. Memory driver card I/O interface

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50 ma (typical) in the high state, and sink up to 50 ma (typical) in the low state.

d. Pins 1 through 240 shall be capable of sending and receiving digital test patterns at rates up to 10 MHZ.

e. All driver circuits shall be short circuit proof, and shall limit output current into a short circuit to the following values:

1. 50 ma minimum to +100 ma maximum for programmable drivers.

f. The dynamic source impedance of driver circuits shall be:

1. 25 ohms maximum for programmable drivers.

g. The comparator circuits of programmable cards shall return a logic 1 when the UUT presents a level in excess of the comparator reference, or a logic 0 when the UUT presents a level below the comparator reference, with the following tolerances:

1. 500 mv for levels up to 5 volts absolute

2. 1 volt for levels between 5 and 15 volts absolute

3. 2 volts for levels in excess of 15 volts absolute

h. Input current (or reference current) from the Unit Under Test logic function (or reference supply) to the programmable comparator circuit shall be equal to the input (or reference supply) voltage minus 2 volts, divided by 4000.

i. The worst case currents[2] drawn by the driver reference voltages (V+ and V-) are :

1. $I+ = -0.5 \text{ ma} + \frac{V+}{4K} + \frac{V+}{R_{load}}$ per Driver in High State

2. $I+ = \frac{V+ + 1V-1}{1.6K}$ per Driver in Low State

NOTE: 2/ When a line is programmed as a UUT output, I+ = I- = 0

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3. I- 0 per Driver in High State

4. $I- = 8.5\ ma + \frac{V+}{1.6K} + \frac{V-}{R_{load}} + \frac{V-}{0.75K}$

per Driver in Low State

j. The programmable driver offset shall be equal to:

$$V\ offset = (0.85 + 0.017\ I_{load}\ (MA)) \pm 10\%$$

k. An external oscillator input (20 MHZ max.) from the interface to the digital word generator shall be provided via card edge connector (Figure 8) A12 J1-A and this input shall have a 56 ohm input impedance with respect to ground. The external oscillator input shall allow operation of the digital word generator/extender from clocks of origin in the UUT or interface device, when programmed for this mode by the user. The external oscillator input will be counted down by a 12-stage programmable counter in the digital word generator, so that the ratio of division between external oscillator input and resulting test clock shall be programmable between values of 2:1 and 8190:1 in binary fashion.

l. An external clock input from interface to the digital word generator shall be provided via card A12 J1-F and shall be designed for operation with TTL or TTL-Schottky driving circuits. The external clock input, when programmed for use, shall be the basic test clock. Maximum frequency of the external clock input shall be 10 MHZ. Input impedance shall be 330 ohms to +5 VDC and 680 ohms to ground.

m. The digital word generator outputs designated CP-1 and CP-2 from card A12 J1-B and J1-C, respectively, shall be identical to internal test clocks, and capable of driving a transmission line terminated by 130 ohms or more. CP-1 is a free running clock. CP-2 is free running except when the handshake mode is programmed in which case CP-2 is synchronized to the external flag signal.

n. There shall be an external flag input from the interface via card A12 J1-D to the digital word generator in the handshake mode. The characteristics of this input shall be equivalent to a 330 ohm resistor to +5 volts DC, and a 680 ohm resistor to ground. The external flag function from the interface (unit under test or adapter) shall be true whenever data has been accepted from the digital word generator and the external device is ready. The external flag shall go false and then true after each command for subsequent commands/ clocks to continue. Logic levels shall be TTL compatible, positive true.

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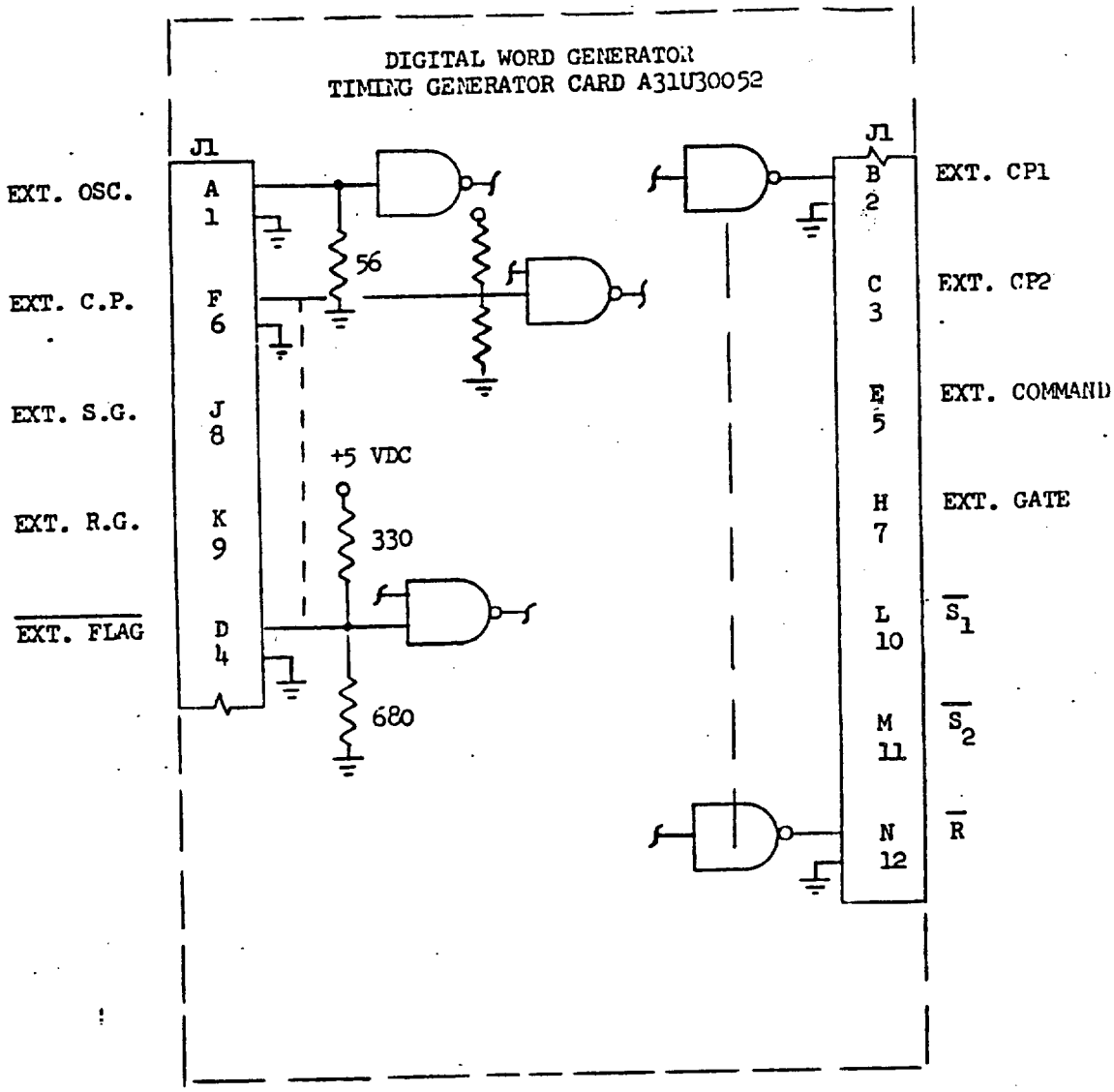


FIGURE 8. Timing Generator card I/O interface.

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o. There shall be an external command signal from the digital word generator to the interface via card A12 J1-E in the handshake mode. The external command signal shall go true whenever the digital word generator presents test data (e.g., a test pattern) to the external device, and shall go false when the external device returns the external flag as acknowledgment. The external command signal shall be a TTL compatible positive true signal, capable of driving a line terminated by 130 ohms or greater.

p. There shall be a gate signal from the digital word generator to the interface via card A12 J1-H. This shall be a TTL compatible positive true signal which goes true with the test initiate function and false when the last stimulus pattern has been terminated. Thus, this function shall be true whenever there is a test in progress. This signal shall drive 130 ohms or more.

q. There shall be an external stimulus gate input from the interface card A12 J1-J. This shall be a TTL compatible true signal which will cause the test to indicate when it goes true and cause the last stimulus pattern to be transmitted when it goes false. Input characteristics, looking into the digital word generator shall be equivalent to a 330 ohm resistor to +5 VDC, and a 680 ohm resistor to ground.

r. There shall be an external response gate input from the interface via card A12 J1-K. This shall be a TTL compatible positive true signal which shall cause the digital word generator to start receiving response patterns when it goes true and will cause the digital word generator to stop recognition of response patterns when it goes false. The input characteristic, looking into the digital word generator, shall be equivalent to a 330 ohm resistor to +5 VDC, and a 680 ohm resistor to ground.

s. There shall be a clock S1 signal from the digital word generator to the interface via card A12 J1-L. This shall be a TTL compatible positive true signal which is identical to the internal digital word generator stimulus strobe, but which occurs only during the times between test initiate and transmission of the last stimulus pattern in a burst. This signal shall drive 130 ohms or more.

t. There shall be a clock S2 signal from the digital word generator to the interface via card A12 J1-M. This signal shall be identical to the clock S1 signal, except that it shall be delayed by 1/2 clock interval with respect to clock S1.

u. There shall be a clock R signal from the digital word generator to the interface via card A12 J1-N.

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This shall be a TTL compatible positive true signal which is identical to the internal digital word generator response strobe, but which occurs only during the times between test initiate and transmission of the last response pattern in a burst. This signal shall drive 130 ohms or more.

v. Slew rate of the programmable driver shall be 300 volts per microsecond, nomincal, at 5 volts.

w. The maximum skew shall be 25 nanoseconds across the full 240 input/output lines, where skew is defined as the worst case time difference between the leading edges of any two (2) stimulus bits in a stimulus pattern as these appear at the DWG output. The maximum skew across any 24 input/output pins of a group of pins associated with a digital word generator circuit card shall be 10 nanoseconds.

3.6 Accessories. Tables I and II respectively list the accessories furnished with and available for use with the DWG.

TABLE I. Accessories available.

NOMENCLATURE	MANUFACTURER, P/N	DESCRIPTION
Connector	Viking 3VH31/1CVS	Mates with Plug-In Card Edge Connector
Hood	Viking, 8030	Hood for above Connector

TABLE II. Accessories furnished.

NOMENCLATURE	MANUFACTURER, P/N	DESCRIPTION
Extender Card	Grumman Aerospace Corp. A31U29009-1	Extends plug-in Cards for Trouble-Shooting
Shorting Plug	Grumman Aerospace Corp. A31U30075-1	Wraparound I/O Signals for Test Purposes

3.7 DWG Programming. Figure 9 is a functional block diagram illustrating signal flow to, from and within the DWG. The signals consist of 16 input lines (instructions and data),

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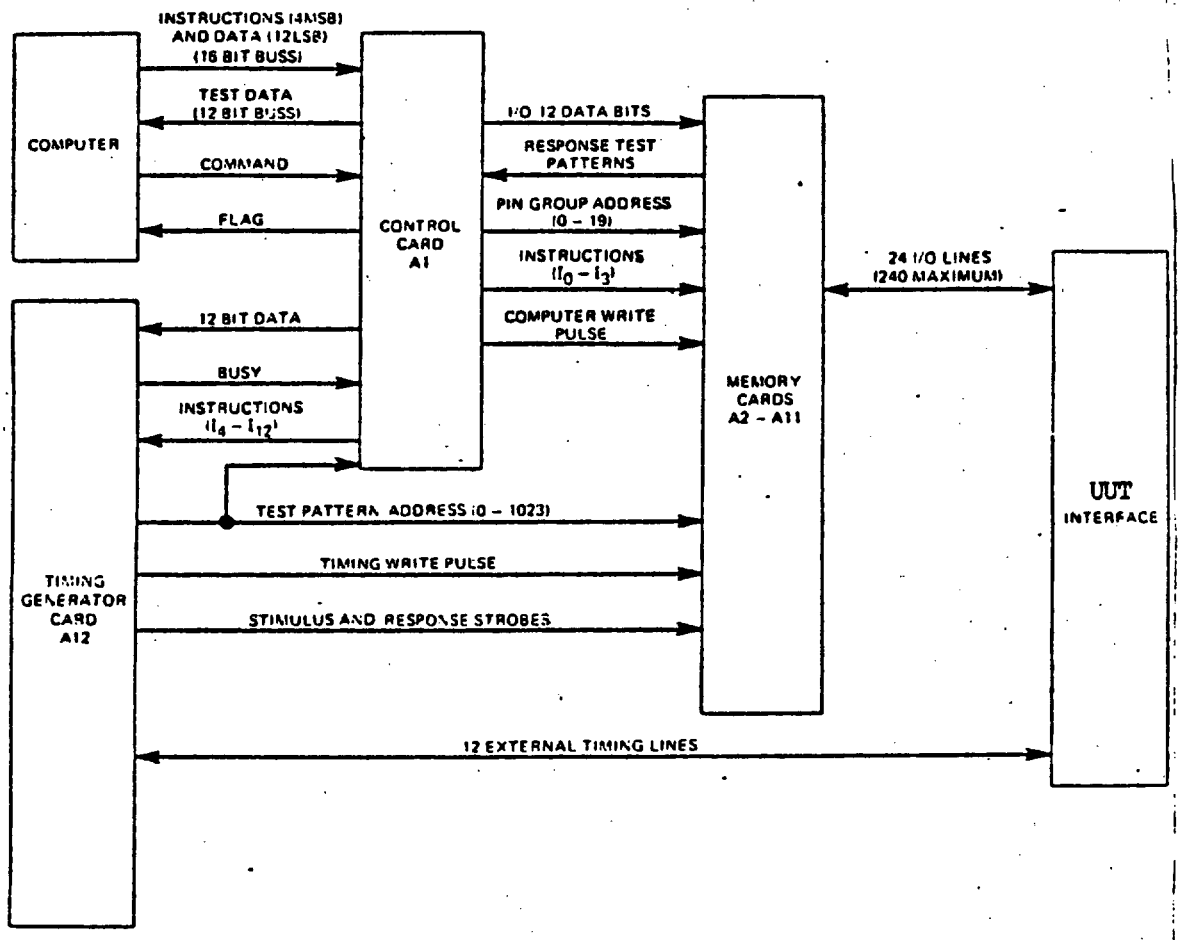


FIGURE 9. DWG functional block diagram.

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12 output lines (data), a command line, and a flag line.

3.7.1 Computer/DWG Operating Mode. The Computer and DWG work in a "handshake mode" (See Figure 10). Operationally, programmed data, from the Computer, appears at the output register of the Computer's I/O interface card. With both the command and flag lines sitting at logic HIGH (inactive state), the action is initiated by the command line going LOW. The DWG detects the negative transition of the command line and responds by setting the flag line LOW. The Computer detects the LOW state of the flag line and reacts by resetting the command line HIGH. Upon completion of the processing cycle, the DWG returns the flag line HIGH (operation complete, DWG ready) and awaits the next instruction. Similarly, test status data, read into the Computer, is controlled the same way. The data is available to the Computer when the flag goes HIGH. The time for completing instruction test cycle is 4 usec nominal, with a range of from 1 usec to 30 seconds (when externally controlled by UUT only).

3.7.1.1 Control Card. The Control Card accepts data and instructions from the Computer in order to control both DWG memory and Timing Generator operation. All data to and from the Computer passes through this card.

3.7.2 Instruction Decoding. Sixteen (16) bit words from the Computer are decoded by the "Instruction Logic and Decoder" on the Control Card. There are basically two (2) types of instructions. The first type is comprised of single word instructions which are the bulk of the repertoire. The second are multi-word instructions of which there are three (3). The functions performed and the formats required for single-word instructions are shown in Figure 11 and multi-word instructions, in Figure 12. Each word sent to the DWG must be accompanied by a COMMAND signal from the Computer. Since the DWG does not store incoming instructions, the Computer is required to hold the data ($\overline{DW}$) during the instruction cycle.

3.7.3 Instruction Cycle. When data is on the lines from the Computer, it is decoded. Action is initiated with the receipt of the COMMAND signal which triggers off the "T" pulse counter. The counter generates four (4) timing pulses (T_0, T_1, T_2, and T_3) each of which is one (1) usec wide. The complete instruction cycle is therefore normally four (4) microseconds. Each of these timing pulses is used to control the individual operations required to complete one (1) cycle. There are two (2) general cases where the cycle is not four (4) usec. The INITIALIZE Instruction requires only one (1) usec. Three (3) instructions (Increment Pattern Counter $-I_4$, Load Pattern Counter $-I_5$, and Test Execute $-I_{11}$) are variable time instructions. They are doing operations which are a fucntion of the frequency of the Timing Generator. At low

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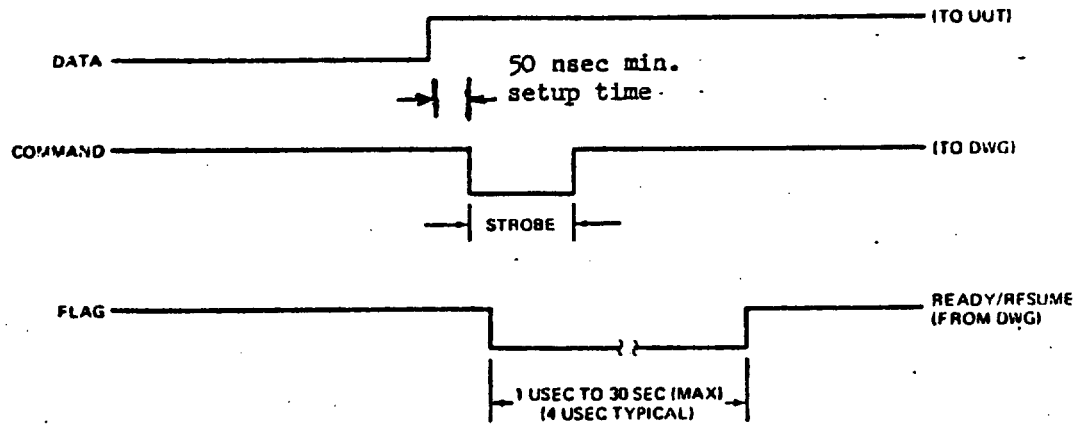


FIGURE 10. DWG/Computer handshake mode.

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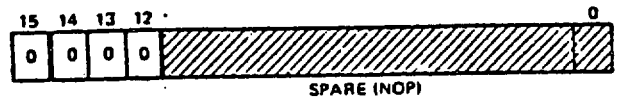
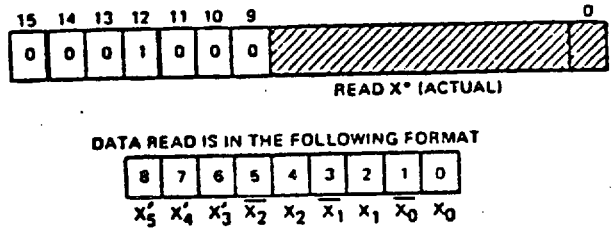
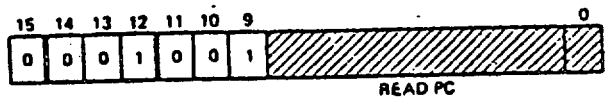
INSTRUCTIONS	FORMAT	FUNCTION
NO OPERATION		NONE
READ PIN GROUP COUNTER		PIN GROUP COUNTER CONTENTS IS PLACED IN DATA OUTPUT REGISTER TO COMPUTER
READ PATTERN COUNTER		PATTERN COUNTER CONTENTS IS PLACED IN DATA OUTPUT REGISTER TO COMPUTER
SET PATTERN COUNTER QUADRANT		SETS QUADRANT OF PATTERN COUNTER TO Y_9Y_8 (1ST, 2ND, 3RD OR 4TH QUADRANT) 256 BITS-PER-QUADRANT
INCREMENT PATTERN COUNTER BY ONE		INCREMENTS VALUE IN PATTERN COUNTER, BY ONE
SINGLE STEP		WHEN DWG IS PROGRAMMED FOR HANDSHAKE MODE, THIS COMMAND IS USED TO STEP THROUGH TEST, ONE PATTERN AT A TIME.

FIGURE 11. Single word instructions.

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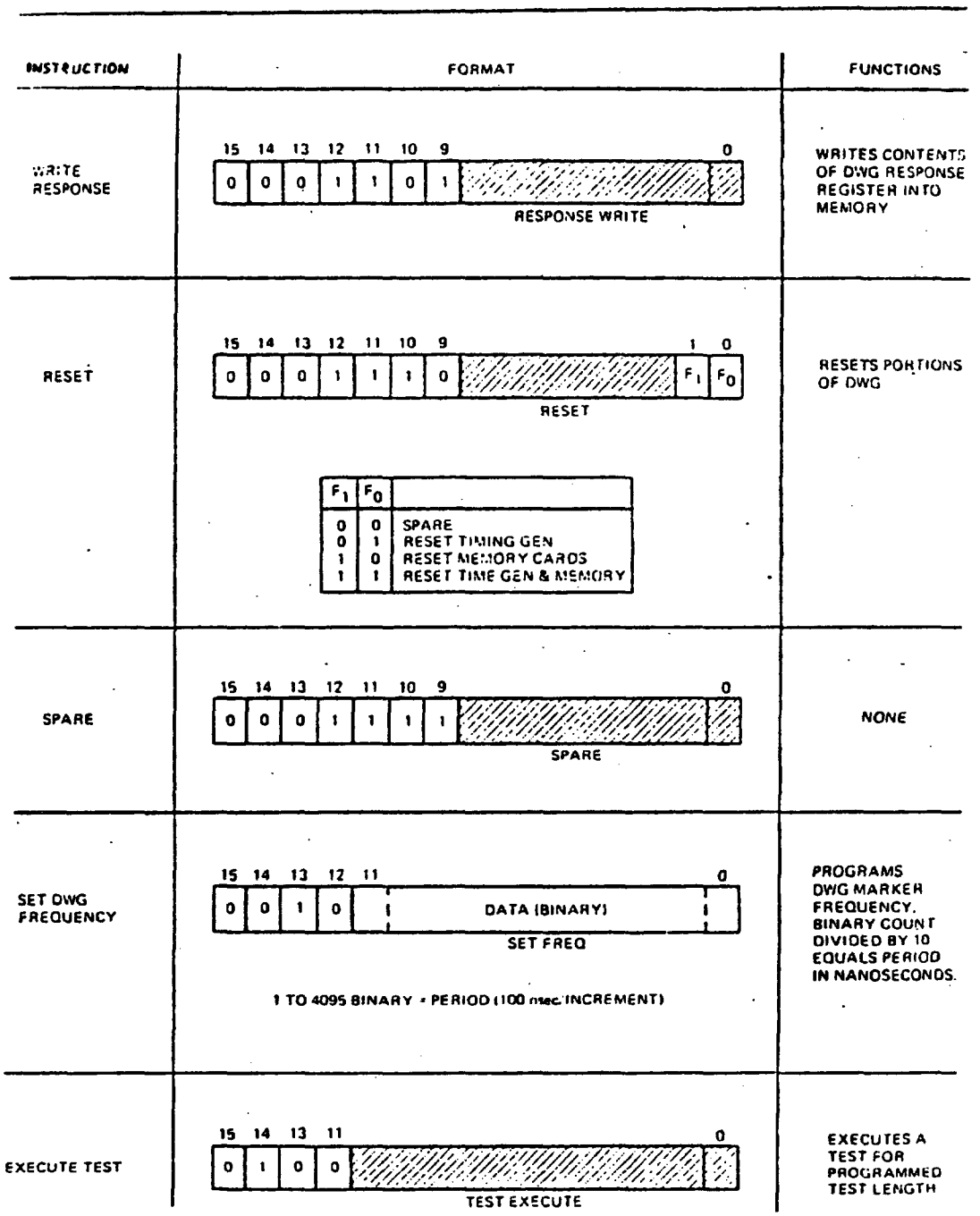


FIGURE 11. Single word instructions - continued.

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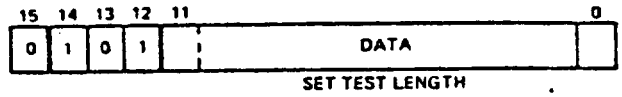
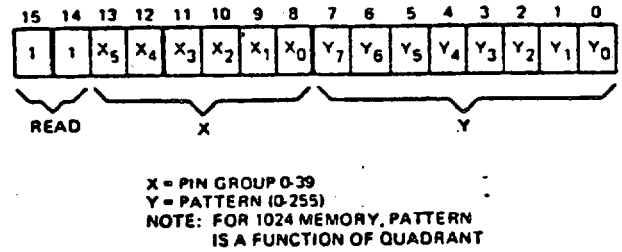
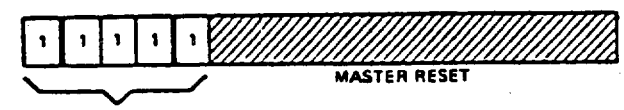
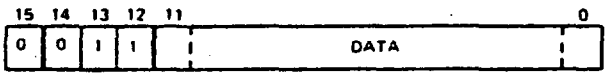
INSTRUCTION	FORMAT	FUNCTION
SET TEST LENGTH		SET THE DWG TEST LENGTH FOR PROGRAMMED VALUE (BINARY 1 TO 4095).
READ MEMORY LOCATION X-Y (ALSO SET PATTERN COUNTER TO Y)		CONTENTS OF MEMORY IS PLACED IN DATA OUTPUT REGISTER TO COMPUTER
INITIALIZE CONTROL CARD		USED TO INITIALIZE AFTER POWER TURN-ON (DOES NOT REQUIRE COMMAND SIGNAL
MASTER DWG RESET		DWG MASTER RESET TO KNOWN STATE (EXCLUDING MEMORIES)

FIGURE 11. Single word instructions - continued.

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INSTRUCTION	FORMAT	FUNCTION
SET UP TIMING GENERATOR CONFIGURATION		SETS UP CONFIGURATION REGISTER IN THE TIMING GENERATOR FOR VARIOUS MODES.

SET TIMING GEN CONFIG

- [11] 1 = S_2 STROBE SELECTED; 0 = S_1 STROBE SELECTED
- [10] 1 = ENABLE S_1 AND S_2; 0 = DISABLE S_1 AND S_2
- [9] 1 = ENABLE R AND W_R; 0 = DISABLE R AND W_R
- [8] 0 = CONTINUOUS MODE; 1 = NOT CONTINUOUS MODE
- [7] 0 = GATED MODE; 1 = NOT GATED MODE
- [6] 0 = S_3 ENABLED; 1 = S_3 NOT ENABLED
- [5] [4] RESPONSE GATE SELECT

5	4	RESPONSE GATE
1	1	STIMULUS GATE 1 DELAYED
1	0	STIMULUS GATE 2 DELAYED
0	1	INTERNAL GATE
0	0	EXTERNAL RESPONSE GATE

- [3] 0 = HANDSHAKE MODE; 1 = NOT HANDSHAKE MODE
- [2] 1 = INTERNAL STIMULUS GATE; 0 = EXTERNAL STIMULUS GATE
- [1] 1 = INTERNAL CLOCK; 0 = EXTERNAL CLOCK
- [0] 1 = INTERNAL OSCILLATOR; 0 = EXTERNAL OSCILLATOR

EXAMPLE: WHEN DATA TO CONFIGURE REGISTER = 1 (SAME AS RESET), THE FOLLOWING CONFIGURATION IS ESTABLISHED:

11	S_2 SELECTED	5,4	SG S1
10	S_2 & S_1 ENABLED	3	NO HANDSHAKE
9	R & W_R ENABLED	2	INT SG = INTERNAL GATE
8	NOT CONTINUOUS MODE	1	INT CP
7	NOT GATED MODE	0	INT OSC
6	S_3 NOT ENABLED		

FIGURE 11. Single word instructions - continued.

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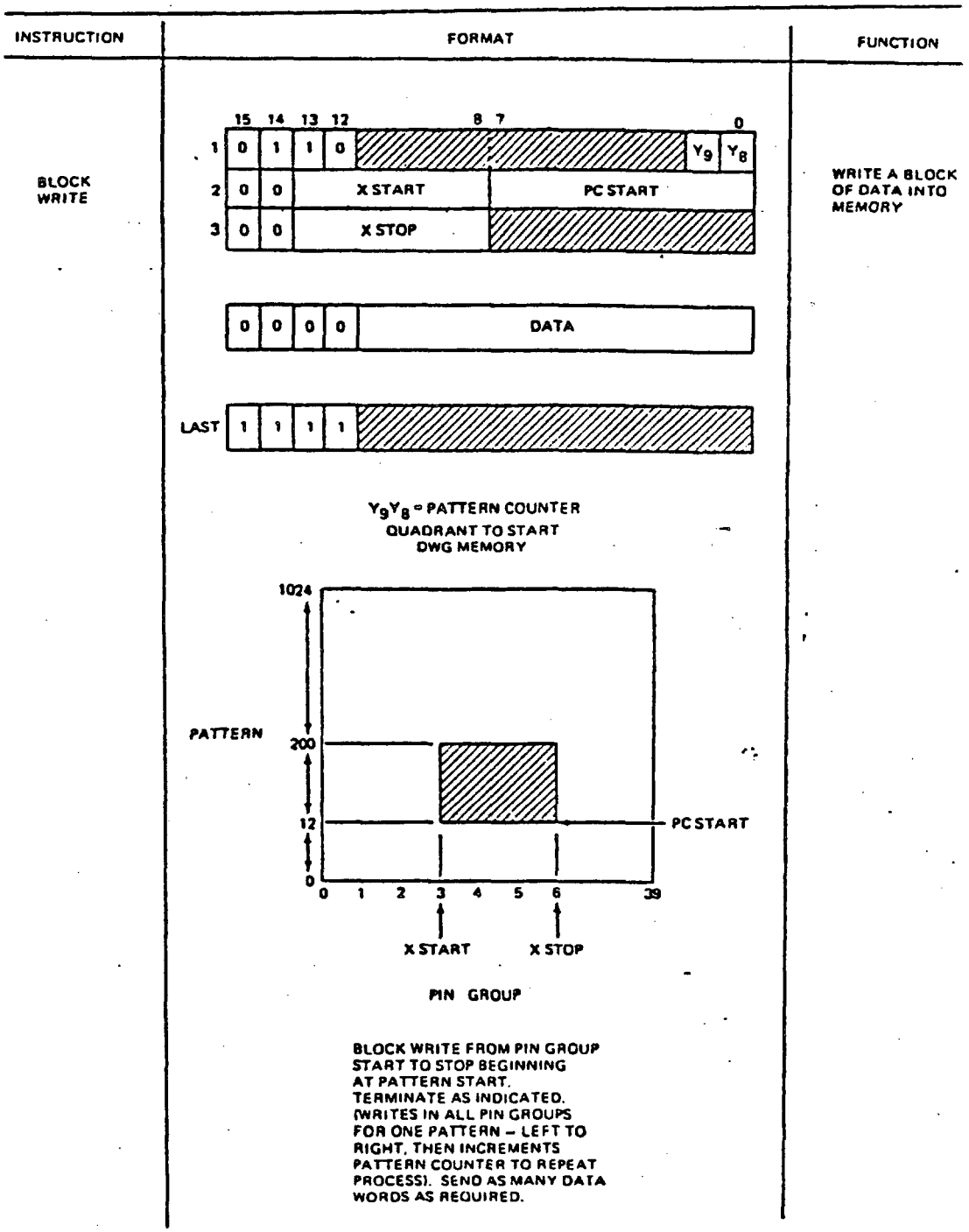


FIGURE 12. Multi-word instructions.

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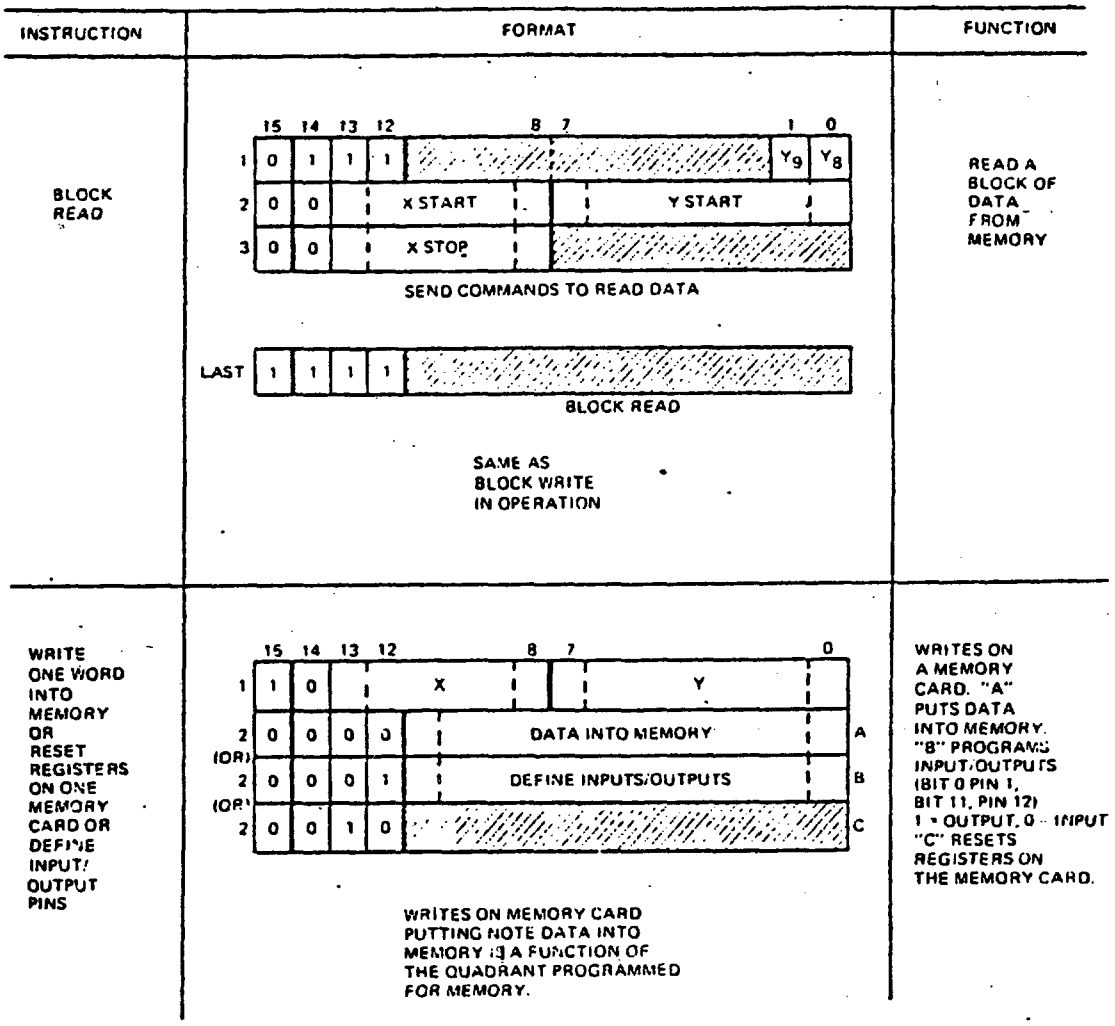


FIGURE 12. Multi-word instructions - continued.

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frequency, these instructions could take more than the nominal. The Busy signal from the Timing Genrator is used by the Control to monitor Timing Generator Status.

3.7.4 Memory Instructions. There are four (4) instructions which address memory cards (I_0 - I_3). These instructions go to all memory cards. The instructions are as follows:

a. I_0 - Defines Input/Outputs

b. I_1 - Reads Memory

c. I_2 - Writes into Memory

d. I_3 - Resets Memory Card Registers (and sets all pins to Outputs) I_0 is a pulse which loads the Input/Output register on the memory card with the data on the write bus ($\overline{DW}$); I_1 gates memory outputs onto the read bus ($\overline{DR}$); I_2 gates the write bus ($\overline{DW}$) into memory; I_3 is a pulse which resets all registers.

3.7.5 Pin Group and Pattern. The addressing of individual memory locations involves controlling "Pin Group" and "Pattern". Memory is divided into 20 pin groups, each group corresponding to 12 pins (group 0 is pins 1 - 12 and group 19 is pins 216 - 240). The Pin Group counter and control logic selects the pin group. The counters output is decoded to produce nine (9) signals (X_0, $\overline{X}_0$, X_1, $\overline{X}_1$, X_2, X_3', X_4', X_5'). The relationship between pin group and signals is shown in Table III.

a. Each slot in the DWG is wired for a particular Pin Group select Address. Therefore, any card can be inserted into any slot. Each Memory Card Decodes the same five (5) pins on every card. For example, the Card in slot XJ4 is for pins 49 to 72 (two (2) pin groups).

PINS	DECODE	PIN GROUP
40-60	$X_2, \overline{X}_1, \overline{X}_0, X_3'$	4
61-72	$X_2, \overline{X}_1, X_0, X_3'$	5

b. To address a particular pattern in memory the Pattern Counter (on the Timing Generator) is used. The Control Card loads the Pattern Counter which, in turn, addresses memory. The only other signal used to control memory is the write signal W_C. W_C is a pulse which writes into memory when I_2 (write instruction) is decoded.

3.7.6 Multi-Word Memory Instructions. There are three

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TABLE III. Pin group signal relationship.

Pin Group		X_2 X_1 X_0	DWG X'_3	DWG X'_4	DWG X'_5	Signal Pins
DWG	0	0 0 0	1			1 - 12
	1	0 0 1	1			13 - 24
	2	0 1 0	1			25 - 36
	3	0 1 1	1			37 - 48
	4	1 0 0	1			49 - 60
	5	1 0 1	1			61 - 72
	6	1 1 0	1			73 - 84
	7	1 1 1	1			85 - 96
	8	0 0 0		1		97 - 108
	9	0 0 1		1		109 - 120
	10	0 1 0		1		121 - 132
	11	0 1 1		1		133 - 144
	12	1 0 0		1		145 - 156
	13	1 0 1		1		157 - 168
	14	1 1 0		1		169 - 180
	15	1 1 1		1		181 - 192
	16	0 0 0			1	193 - 204
	17	0 0 1			1	205 - 216
	18	0 1 0			1	217 - 228
	19	0 1 1			1	229 - 240

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(3) multi-word instructions to control memory. The Control Card has logic for each of these which remember which instruction, in the sequence, is being executed. This logic also disables the decoder so that the DWG does not misinterpret subsequent words incorrectly. The three (3) logic blocks are "Write Control", "Block Write Control" and "Block Read Control". The X counter also works in tandem with the Block Read and Block Write by counting from the Start to Stop pin group, then resetting and repeating the process.

3.7.7 Timing Generator Instructions. The Control Card sends nine (9) commands to the Timing Generator. These nine (9) commands control all facets of Timing Generator Operation. The commands are as follows:

a. I_4 - Increment Pattern Counter by one (1)

b. I_5 - Load Pattern Counter (from $\overline{DW}$ BUS)

c. I_6 - Load Configuration Register (from $\overline{DW}$ BUS)

d. I_7 - Reset

e. I_8 - Load Pattern Counter Quadrant (from $\overline{DW}$ BUS)

f. I_9 - Load Frequency Counter (from $\overline{DW}$ BUS)

g. I_{10}- Load Test Length Counter (from $\overline{DW}$ BUS)

h. I_{11}- Test Exeucte

i. I_{12}- Single Step

3.7.8 Output Data to Computer. A twelve (12) bit register on the Control Card is used to hold data requested by the Computer. Data may be read from the following:

a. Memory - via Read UBS ($\overline{DR}$) (DWG and Extender).

b. Pattern Counter - via Pattern Counter Bus.

c. Pin Group Select - Internal to Control Card.

3.8 Mechanical Interface

3.8.1 Design and Construction. The DWG is packaged in one (1) drawer and designed for rack mounting (see Figure 13). Chassis slides are required for rack counting to provide equipment support and access to plug-in circuit cards and internal componenets. Figure 13 provides the necessary outline dimensions, mounting and mating connector data. Ventilation is provided via an exhaust fan located at the rear of the DWG drawer. A minimum of 3/4" blank spacer is required beneath the DWG drawer to allow for air intake via the bottom panel.

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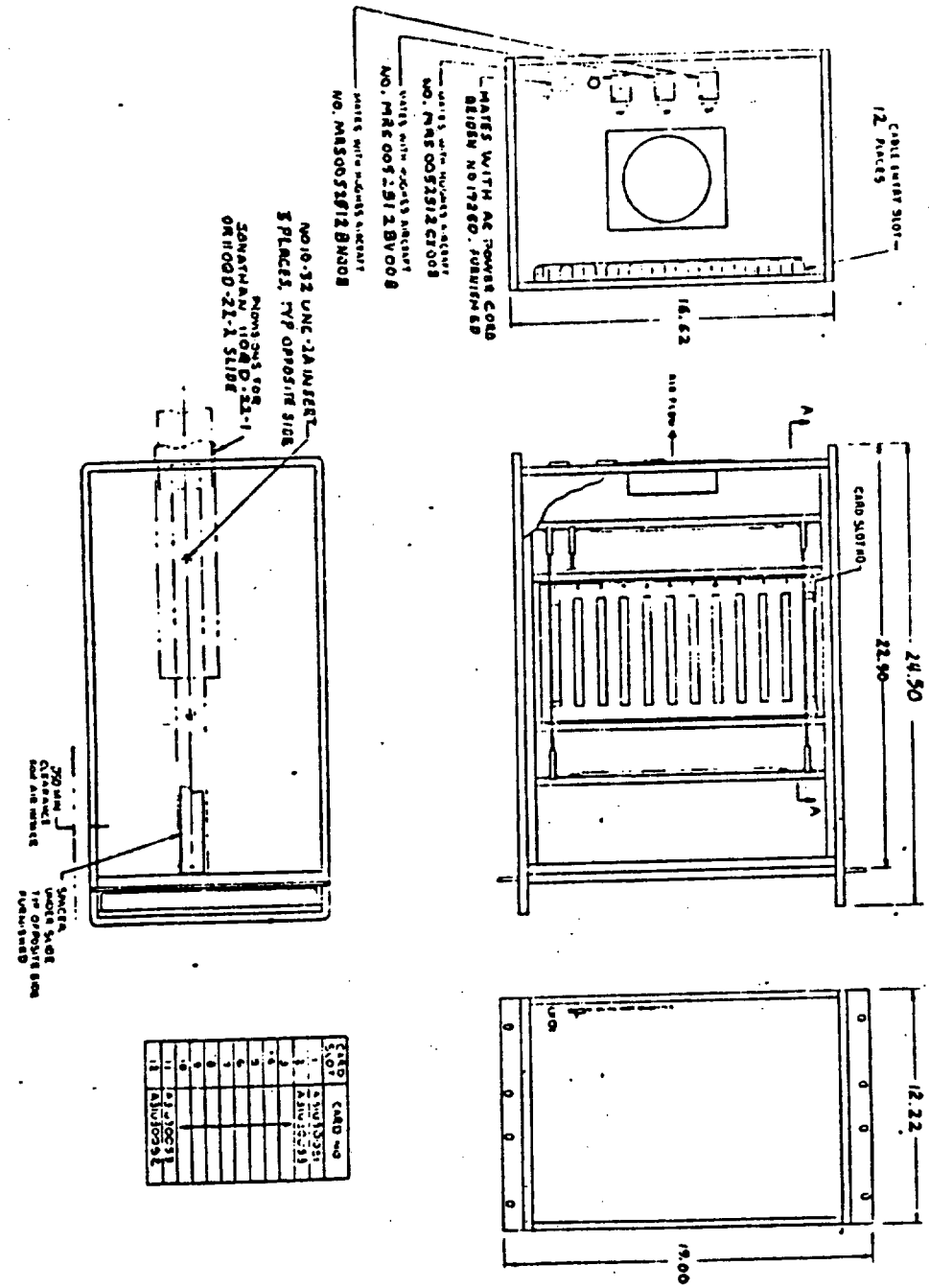


FIGURE 13. Outline drawing.

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3.8.2 Service Conditions. No fixed part or assembly shall become undesirably free or sluggish in operation; no moveable part or control shall be shifted in setting, position or adjustment; and there shall be no performance degradation during operation after storage or use of the System in the ambient conditions specified in the following paragraphs.

3.8.2.1 Non-Operating Temperature. The equipment shall withstand -20°C to +65°C.

3.8.2.2 Non-Operating Altitude. The equipment shall withstand altitudes from 0 to 50,000 feet MSL (30.0 to 3.4 in. Hg.).

3.8.2.3 Operating Temperature. The equipment shall withstand 0°C to 55°C.

3.8.2.4 Operating Altitude. The equipment shall withstand altitudes from 0 to 5,000 feet MSL (30.0 to 16.8 in. Hg.).

3.8.2.5 Humidity. The equipment shall withstand the effects of ambient air relative humidity up to 80% (non-condensing) within the specified operating and non-operating temperature ranges, as appropriate.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection Conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified, in paragraph 3.8.2 contained herein.

4.3 Methods of Inspection and Test. Inspection and test of the DWG shall be made as an integral part of the HTS in accordance with the Acceptance Test Requirements cited in the Hybrid Test System Specification MIL-H-85330.

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5. PACKAGING

5.1 Product Identification and Marking. The DWG, its assemblies and subassemblies shall be marked in accordance with source control drawings or manufacturer's drawings, as defined in the Hybrid Test System Specification MIL-H-85330.

5.2 Packaging Requirements. All major units and parts of the DWG shall be preserved, packaged and marked in accordance with the Hybrid Test System Specification MIL-H-85330.

6. NOTES AND CONCLUDING MATERIAL

6.1 Intended Use. The DWG is intended for use in a computer controlled test system for functional and diagnostic test of digital circuits of Units Under Test at both shore and sea based facilities.

Preparing Activity:

Navy - AS

(Project Number 6625-N764)

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APPENDIX

DESCRIPTION AND THEORY OF OPERATION

10. SCOPE

10.1 Scope. This appendix describes the DWG and its theory of operation.

20. DESCRIPTION

20.1 Overview. The DWG is packaged in one (1) drawer which can accommodate up to 240 test pins, where each pin may be alternately a driver or a receiver.

20.2 Theory of Operation. The DWG is intended to form a part of an automatic, computer controlled test system. It provides up to 240 bidirectional input/output lines, as illustrated in Figure 14 for sending or receiving digital pattern data. Each input/output line has a dedicated high speed memory of 256 bits.

20.3 Drivers and Receivers. The drivers and receivers on each input/output line have programmable logic levels and must be supplied with three (3) external voltages. Two (2) voltages are used for the driver (one (1) is for the driver high logic level and one (1) is for the driver low logic level), and a third voltage is used for the receiver to establish a logic level threshold for incoming data. Any value above the reference is interpreted as a logic 1 while any value below the reference is interpreted as a logic 0.

20.4 Typical DWG Test Mode. In a typical test mode, the DWG memory is loaded with stimulus pattern data (up to limit of memory size). The patterns are then sent to the Unit Under Test (UUT) in sequence and UUT output data is captured by the DWG and stored for evaluation. In using the DWG for UUT testing, many programmable modes are available.

20.5 Inputs and Outputs. There are several basic concepts concerning DWG operation which must be understood in order to write test programs. First, is the fact that the DWG supplies up to 240 bidirectional input/output lines for sending and receiving digital pattern data. A single pattern is defined as the I/O sequence which appears on these lines simultaneously. A stimulus pattern to the UUT is defined in the programming as an INPUT pattern while a response pattern from the UUT is defined as an OUTPUT pattern. The first step in using the DWG is to estalbish which of the lines are to be used as INPUT's and which are to be used as OUTPUT's. The

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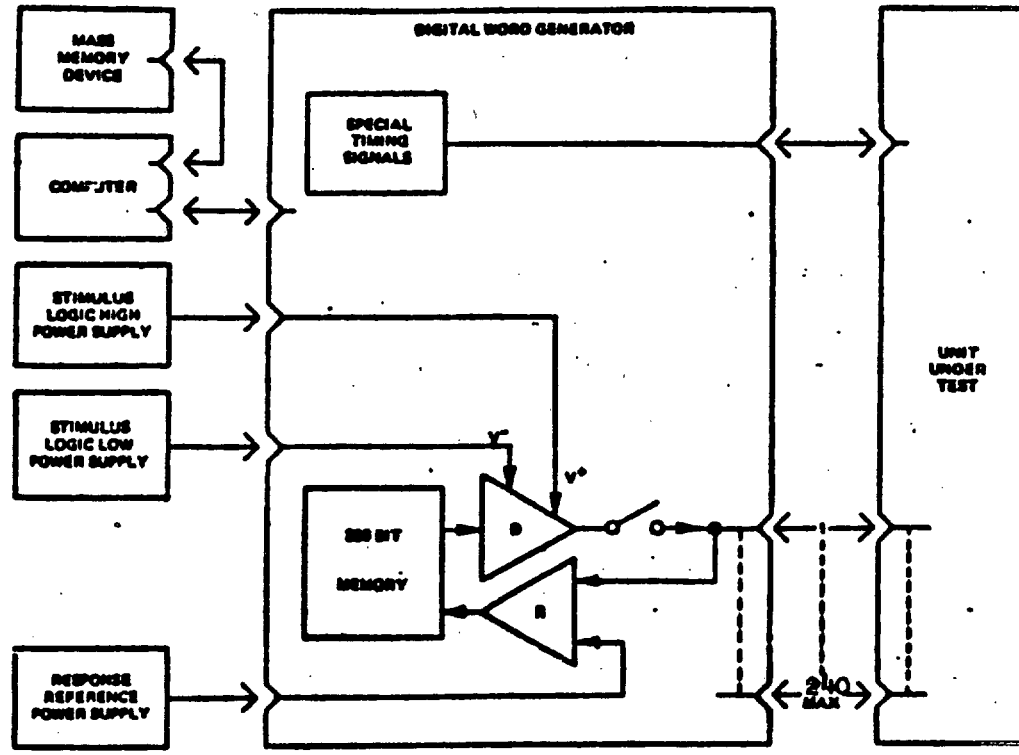


FIGURE 14. Simplified test setup diagram.

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next step would be to program the desired Driver and Receiver logic levels. As illustrated in Figure 14 the levels are provided by power supplies external to the DWG.

20.6 UUT Testing

20.6.1 Having defined INPUT's and OUTPUT's, it is now possible to perform UUT tests. A test sequence would involve sending an INPUT pattern to a UUT and then sampling a sesultant OUTPUT pattern and comparing it with predicted results to ascertain UUT GO/NO-GO status. The DWG hardware system sends and receives the patterns, while the software system does the comparisons of actual UUT OUTPUT's with predicted or Good Machine Responses (GMRs). Patterns to be sent to a UUT are first stored in DWG memory. They are then transmitted while simultaneously the UUT response patterns are captured and stored in DWG memory. From there, they can be brought back to the computer for analysis.

20.6.2 During the actual transmission and reception of patterns, the DWG timing generator controls timing functions. Both internal and external synchronization features are provided. The standard test mode would be to use the DWG on internal synchronization. In this mode, two (20 fundamental parameters should be considered. The first of these is the desired time interval from INPUT pattern transmittal to a resultant OUTPUT pattern sample. This time interval is programmable as a multiple of the main DWG frequency. The second is the main frequency at which successive patterns are to be transferred (i.e. time interval between two (2) INPUT patterns). A sequence of patterns can be transferred to the UUT at some desired frequency up to the limit of memory capacity. When memory is exhausted, testing is halted while additional data is retrieved from the computer. The sample time between INPUT AND OUTPUT, the main frequency, and the number of patterns to be bursted are all programmable.

20.6.3 The timing and synchronization of the DWG is controlled by a master oscillator. From it, three (3) stimulus strobes (S1, S2 and S3) and one (1) response strobe (R) are developed. The frequency of these strobes is equal to the programmed DWG frequency. These strobes are active during actual test execution. Each DWG input/output line is synchronized to these strobes. Normally, INPUT's are controlled by S1 and S2 and OUTPUT's by R.

20.6.4 Timing relationships between the basic period and S1, S2, S3 and R is indicated in Figure 15. Strobe S3 is identical to S2 except that it is independently controlled. The programming period determines the frequency at which data is transmitted from DWG memory. Stimulus data is transmitted on the leading negative edges of either S1 or S2, depending

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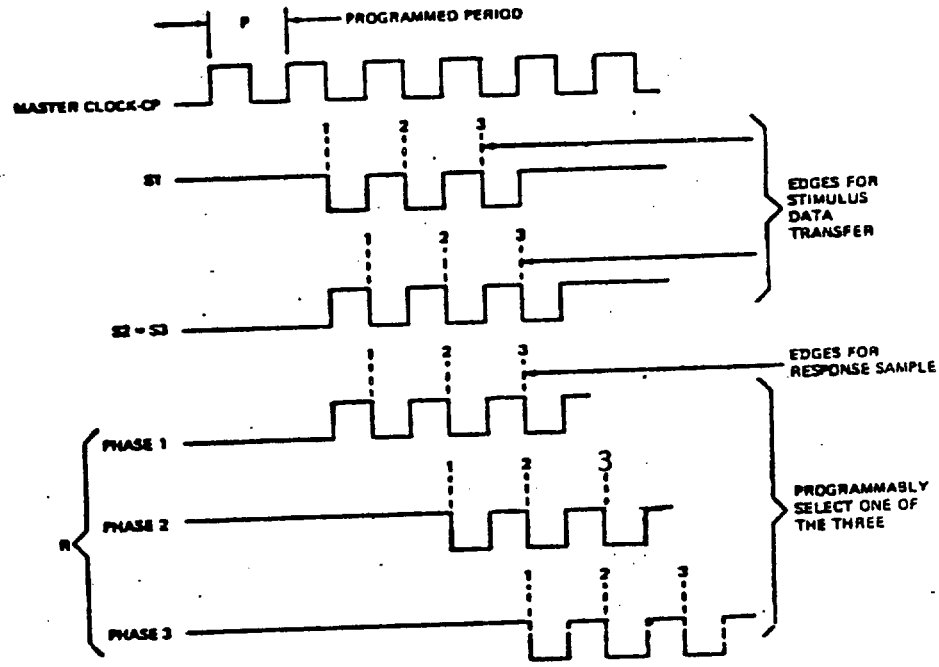


FIGURE 15. Timing relationships.

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upon which is selected in the programming. Response data is captured on the leading negative edges of R. Strobe R can be programmed for one (1) of three (3) phases as shown. By selecting combinations of S1, S2 and R, the time interval between stimulus transmittal and response sample can be controlled as a multiple of the master frequency. For example, if S1 and R - phase 1 were selected, the time interval would be equal to one-half (0.5) the period. Likewise, other combinations are possible from one-half (0.5) to two (2) periods. The period is also programmable from 0.1 usec to 409.5 usec, so that the time interval can be controlled with a 50 nsec resolution.

20.6.5 Having set up the timing, several operational modes are programmable. They are as follows:

a. Normal - Processes patterns, sequentially, from DWG memory. The number of patterns is programmable (test length).

b. Continuous - Recycles continuously the programmed number of patterns.

c. Gated - Recycles every other time frame the same series of patterns.

d. Handshake - Explained in paragraph 20.6.6.

20.6.6 Of the four (4) modes, the Handshake mode requires explanation. It is a mode which, after a stimulus pattern, is sent from the DWG. The DWG goes into a "wait" condition before sampling the response. It will sample responses, when directed to do so either programmably or by the UUT. Upon execution of this command, the DWG will sample the response and then send the next stimulus. It will then go to a "wait" condition. The UUT can command the DWG using an external control fucntion (FLAG line).

20.6.7 For the flag function, a second DWG external line (COMMAND line) is also involved (see Figure 16). The COMMAND line goes positive coincident with stimulus data transfer. The DWG waits until a FLAG pulse is returned from the UUT. On its positive edge, the DWG resumes operation and samples the response. The DWG then sends the next pattern and COMMAND and again, waits for the FLAG.

20.6.8 The remaining modes involve external DWG timing functions. For this purpose, ten (10) external lines are provided (in addition to the COMMAND and FLAG lines). They are as follows:

a. DWG Inputs

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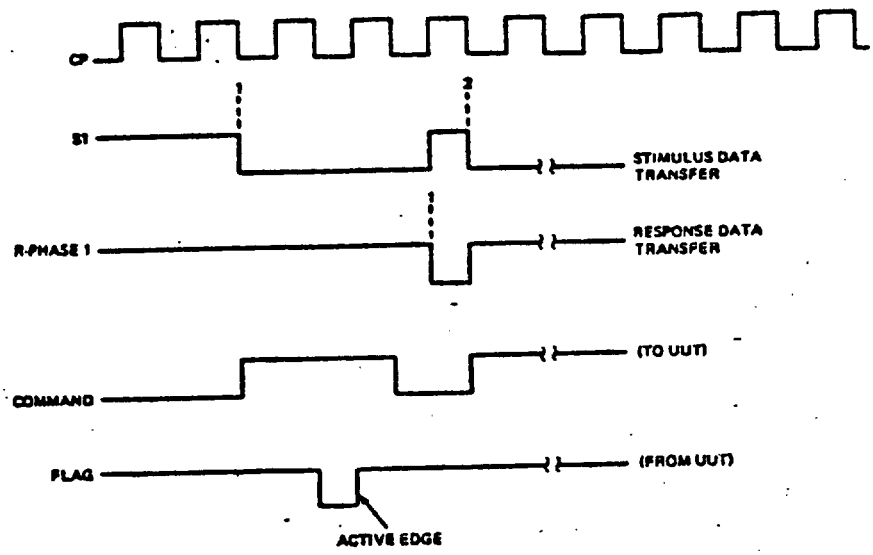


FIGURE 16. Handshake mode.

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1. External Oscillator
2. External Clock
3. External Stimulus Gate
4. External Response Gate

b. DWG Outputs

1. Clock CP_1
2. Clock CP_2
3. Strobe S1
4. Strobe S2
5. Strobe R
6. GATE

20.6.9 The EXTERNAL Oscillator permits the DWG to operate from an external oscillator in place of its own internal oscillator. The external oscillator is counted down in the DWG to produce the master clock. The EXTERNAL CLOCK permits the DWG to operate directly off an externally provided clock. The EXTERNAL STIMULUS GATE, when true, directs the DWG to send stimulus patterns. The EXTERNAL RESPONSE GATE does the same for response patterns. The remaining external functions are all DWG signal outputs. Clocks CP_1 and CP_2 are the DWG's internal clocks. CP_1 and CP_2 are identical and are both free-running. The only difference is that CP_2 is not free-running in the HANDSHAKE mode. Rather, it is synchronized to the external FLAG signal. Strobes S1, S2 and R are identical to the internal DWG stimulus and response strobes. The last output is the GATE which is true during stimulus pattern transmission.

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